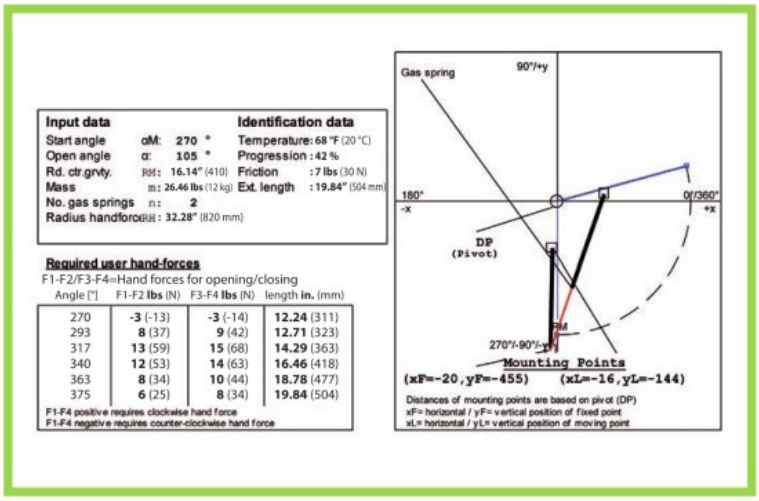
Calculation

To obtain the ideal selection to give the optimum operation for a gas spring it is important to identify the following points:

- gas spring size
- required gas spring stroke
- mounting points on flap and frame
- extended length of the gas spring
- required extension force
- hand forces throughout the complete movement on the flap

With our **free calculation service** you can eliminate the time-consuming calculation and send us your details by fax or e-mail. Just complete the information shown on the calculation formula page number 6. Please attach a sketch of your application (a simple hand sketch is sufficient) in side view. Our application engineers will determine the optimum gas springs and mounting points and calculate the ideal situation to satisfy your requirements.



You will receive a quotation showing the opening and closing forces and our recommended mounting points to suit your application.

"Calculation offer with all required details for assembly!"



Safety Instructions

Gas springs are filled with nitrogen gas. Nitrogen is an inert gas that does not burn or explode and is not poisonous.
Please note!: the internal pressure of gas springs can be up to 4,350 psi (300 bar). Do not attempt to open or modify them.

ACE gas springs will operate in surrounding temperatures from -4 °F (-20 °C) to 176 °F (+80 °C). We can equip our springs with special seals to withstand temperatures as low as -49 °F (-45 °C) or as high as 392 °F (+200 °C). Gas springs should not be placed over heat or in open fire!

Disposal/Recycling: Gas Springs consist mostly of metal and the metal could be recycled, but first the gas pressure must be removed. Please ask for our disposal recommendations which advise how to depressurize the gas springs and make them safe to recycle.

All gas springs are marked with the part number, the production date and a warning sign "Do not open high pressure". We are not responsible for any damages of any kind that arises due to goods that are not marked accordingly.

Gas springs should be installed with the piston rod downwards. This position ensures best damping quality. **Only ACE gas springs include an integrated grease chamber which allows for alternative mounting opportunities.**

Gas springs should not be exposed to tilting or side load forces during operation or whilst static (this can cause bending of the piston rod or early wear).

Gas springs are maintenance-free. **Do not grease or oil the piston rod.**

The piston rod must not be painted and should be protected against shocks, scratches and dirt. The cylinder should not be deformed as such damage would destroy the sealing system.

ACE gas springs can be stored in any position. Pressure lost through long storage is not to be expected. There may be a sticking effect the first time you compress a spring. This may require a higher initial force to operate the gas spring for the first time (initial breakaway force).

The tolerance for the installation length is generally deemed to be ±.08" (± 2 mm). If very high demands are placed on durability and stability, please avoid the combination of small diameter + long stroke + high force.

The force tolerance is ±7%.